

# IRSpectra-Bench and IReXP: candidate recall, not verification, limits LLM elucidation from real experimental IR and NMR

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We release *IReXP*, the largest openly redistributable collection of experimental infrared band lists (121,233 records; 43,060 structure-linked; 33,201 full IR + <sup>1</sup>H + <sup>13</sup>C + structure quadruples), and *IRSpectra-Bench*, a blind mechanically scored benchmark of 194 compounds drawn from it. On *IRSpectra-Bench*, given the molecular formula and peak lists exactly as reported in open-access papers, a frontier large language model recovers the correct constitution for 28% (top-1; 95% CI 22–35), or 15% once reweighted to the corpus composition — far below the near-100% implied by curated demonstrations. The bottleneck is candidate proposal, not verification: the true structure enters the pool for only 34% of compounds, and where it does, training-free forward-verification selects it 89% of the time (58/65; 81% on the 37 compounds where a ranker had a choice). Recovered from published top-*k* figures, the same split carries 68–83% of the accuracy collapse when three systems move from curated or simulated data to real heterogeneous spectra. Generating wider lifts recall 32% → 42% and top-1 23% → 30% on 60 compounds; verification itself moves whole-benchmark top-1 only 28% → 30%. Masking the spectra drops top-1 from 23% to 5%, and Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol all verify better than they generate. All predictions and code are released.

## 1. Introduction

Determining structure from spectra is central to synthetic and analytical chemistry. Machine learning uses encoder-decoder models on paired corpora; *Spectro*, for one, learns <sup>1</sup>H/<sup>13</sup>C/IR → SELFIES from 6,833 molecules<sup>1</sup>. General-purpose LLMs are now reported off-the-shelf: a non-peer-reviewed 2026 industrial white paper found Claude Opus matching or beating commercial NMR-prediction software forward (structure → spectrum, ±0.08 ppm <sup>1</sup>H) and “recovering all eight simpler structures on every attempt” in the inverse<sup>2</sup>. We treat those numbers as a claim to test, not an established baseline.

That evaluation is narrow: 15 inverse problems on curated single-ring or two-fragment molecules, NMR only, seven given the *starting-material structure* as a hint, and “recovery” scored leniently over three runs and three ranked candidates. The chemist’s question — *take an arbitrary experimental spectrum from a paper and recover the structure* — remains open. It needs open experimental data at scale, a blind benchmark others can report against, and honest accounting of which stage of elucidation binds.

We provide all three. *IReXP* (§2) is, to our knowledge, the largest openly *redistributable* collection of experimental IR band lists (121,233 records; 43,060 structure-linked; 33,201 IR + <sup>1</sup>H + <sup>13</sup>C + structure); view-only libraries such as SDBS hold more structure-linked *spectra* but cannot be redistributed (§2.1). *IRSpectra-Bench* (§3) is the blind, mechanically scored benchmark built from it — 194 compounds, complexity-stratified, multimodal IR + <sup>1</sup>H + <sup>13</sup>C. Concurrent 2026 suites bound related tasks (MolQuest scores 530 post-2025 compounds by exact canonical SMILES<sup>3</sup>; SpecX<sup>4</sup>; NMRGym<sup>5</sup>); ours is an openly redistributable suite on literature-reported peak lists with a recall/verification decomposition measured on Claude, replicated across three other model families (§4.7),

with a within-compound solver-methodology control (§4). Ground truth is literature structures resolved deterministically (OPSIN/RDKit) and checked mechanically (560/560 bands on a seed-fixed sample, §2.3; expert-chemist review of elucidation outputs is formally deferred, §8); no LLM curates labels or scores predictions.

Forward-verification elucidation (§5) turns the model’s *strong* direction (forward prediction) against its *weak* one (inverse regiochemistry); the loop is prior art (§1.1), the decomposition ours, run training-free on every compound. The finding is sharp asymmetry: given a candidate set containing it, the model *verifies* the correct structure 89% of the time yet *proposes* it for only 34%, so generation binds the result (Fig. 1). Gain is bounded (top-1 28% → 30% on n=194; 23% → 30% on the 60-compound arm where wide generation was also tested) and, by our own measurements, cannot exceed a recall/precision ceiling without sharper verification or 2D-NMR data.

We contribute *IReXP* and *IRSpectra-Bench* — open peak lists plus decomposable recall/verification metrics others can report — measured on every compound (34% proposed; 89% selected once proposed), with a four-vendor replication of that split. The paper fills an accounting and infrastructure gap: it diagnoses where elucidation binds; it does not solve it.

Solver and verifier (Fig. S1) are LLM agents on a consumer subscription, with no fine-tuning and no API spend. Inference is not exactly reproducible — no pinned snapshot, temperature or seed (§9) — but scoring is: predictions, ground truth and scorers are released, and every training-free number regenerates from them. The HOSE lookup, GNN (§5.4) and trained generator (§5.6) are fenced complements.

### 1.1 Related work

**Trained spectra → structure models.** The dominant line trains sequence or graph decoders on paired corpora — *Spec*-



**Fig. 1.** Diagnosis on IRSpectra-Bench (n=194): generation recall, not verification, is the bottleneck. Where the true structure is never proposed, no ranker can recover it; where it is proposed, verification usually selects it. End-to-end top-1 is the product of those two rates, which have different denominators and are not differenced.

*tro*<sup>1</sup>, multitask 1D-NMR models<sup>6</sup>, NMRTrans/NMRSpec<sup>7</sup>, *NMIRacle* (IR + <sup>1</sup>H + <sup>13</sup>C)<sup>8</sup> — accurate in-distribution but retrained per modality. IRExp (§2.3) complements NMRSpec with redistributable IR band lists; neither corpus alone is multimodal. Upstream IR work from our group (*vIR-OLO*<sup>9</sup>, *j-IR-vis*<sup>10</sup>) identifies functional groups but does not generate candidate structures.

**LLMs as elucidators.** LLM agents already plan syntheses (ChemCrow<sup>11</sup>, Coscientist<sup>12</sup>) and call elucidation routines; we measure what one returns. Off-the-shelf and multimodal LLMs<sup>2,13–15</sup>, puzzle benchmarks (MolPuzzle<sup>16</sup>), and agentic IR interpreters (IR-Agent<sup>17</sup>; Priessner *et al.*<sup>18</sup>) improve *how* a model reads spectra or re-ranks a fixed pool; we ask which stage binds once it has read them.

Most benchmarks use simulated or single-instrument spectra<sup>1,8,19,20</sup>; IRSpectra-Bench uses literature-reported peak lists across thousands of laboratories. Cross-paper numbers swing with harness: GPT-4o scores 1.4% on MolPuzzle<sup>16</sup>, 27.8% with chain-of-thought, 57.8% with tree-search<sup>15</sup>. We fix one pre-registered RDKit-InChIKey protocol with bootstrap CIs.

**Forward verification and CASE.** Matching predicted to observed shifts — DP4<sup>21,22</sup>, NMR crystallography<sup>23,24</sup>, CASE<sup>25,26</sup> — is easier than inverse generation. We swap DFT for a forward LLM. CASE enumerators have exhaustive recall by construction; the LLM literature almost never reports whether the true structure was proposed at all (§6).

**Prior loops and concurrent work.** *NMR-Solver*<sup>27</sup> runs the same generate-and-verify loop without an LLM (52.9% top-1 on literature <sup>1</sup>H/<sup>13</sup>C, formula supplied; 31.6% with stereochemistry). We decompose error rather than beat that score. *NMRAgent*<sup>28</sup> is a complementary best-effort system. Kliuev *et al.* place the best of six LLMs at 24% on 105 peak lists<sup>29</sup>. Espejo Morales *et al.* frame agentic search with processing tools and hard checks: 80.9% on education spectra, 20.6% on 34 industrial samples<sup>30</sup> — they ask which architecture; we ask which stage binds. Their inputs are raw instrument files; ours are published reports (§8). Priessner *et al.* already state that re-ranking cannot recover a missing candidate<sup>18</sup>. We add measurement at scale: the split on 194 compounds across four model families and four verifiers, recovered from published top-*k* figures (§6).

## 2. The IRExp dataset

### 2.1 Motivation

An IRExp record is a *band list* — peak positions in cm<sup>-1</sup> transcribed by an author into a paper’s text — with co-reported <sup>1</sup>H/<sup>13</sup>C shift lists and, where resolvable, the 2D structure. IRExp contains no absorbance traces. That is the published form a language model consumes, but a different object from a digitised spectrum, and the two should not be counted against one another.

Among *digitised spectra*, free downloads run to the NIST WebBook<sup>31</sup> (≈16k) and the Chemotion ELN deposit<sup>32</sup> (≈2k). AIST SDBS<sup>33</sup> is larger (≈54k FT-IR, all structure-linked) but *view-only* (50 spectra/day, no bulk export), so IRExp does not compete there: SDBS alone holds more structure-linked spectra than IRExp holds structure-linked band lists. Among text-derived *band lists* IRExp is the largest *openly redistributable* collection by record count — deliberately scoped to that object type.

### 2.2 Construction

Experimental sections report per-compound IR band lists with co-reported <sup>1</sup>H/<sup>13</sup>C NMR in a stable textual convention. Mining it is mature<sup>34</sup>; we add a pipeline specialised to the *IR band list*, the modality those tools passed over.

- **Discovery.** Open-access literature is harvested in bulk from the PMC Open-Access Subset<sup>35</sup> and the Chemotion FT-IR deposit (RADAR4Chem).
- **Extraction.** A deterministic parser segments experimental text per compound and extracts IR wavenumbers and <sup>1</sup>H/<sup>13</sup>C shifts, gated against scan-range artefacts and prose false-positives.
- **Resolution.** IUPAC names go to SMILES with OPSIN<sup>36</sup> and RDKit<sup>37</sup> canonicalisation (InChIKey, SELFIES<sup>38</sup>), with a PubChem<sup>39</sup> fallback for trivial names; handling open-access label conventions and narrative artefacts raised structure coverage from 24% to 35%.

### 2.3 Contents and licensing

Table 1 summarises contents and provenance.

The pools differ: PMC records are author-transcribed (median 9 bands), Chemotion records peak-picked from deposited spectra (median 39); users training on band density should treat them apart. `scripts/split_license_pools.py` separates them on `source_doi` and stamps each record’s licence.

**Extraction fidelity is measured.** On a seed-fixed random sample of 60 PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`) we re-fetched

Table 1. IRExp dataset contents and provenance.

| field                                                                     | value          |
|---------------------------------------------------------------------------|----------------|
| experimental IR band-list records                                         | <b>121,233</b> |
| ...co-reporting $^1\text{H}$ and/or $^{13}\text{C}$ NMR                   | 87,075 (72%)   |
| ...with a resolved 2D structure                                           | 43,060 (35.5%) |
| ...of these, with $^1\text{H}$ and/or $^{13}\text{C}$ NMR                 | 40,702         |
| ...of these, with both $^1\text{H}$ and $^{13}\text{C}$ (full quadruples) | 33,201         |
| by provenance: author-transcribed from PMC-OA text (CC-BY)                | 119,345        |
| by provenance: peak-picked from Chemotion ELN deposits (CC-BY-SA)         | 1,888          |

each article and checked every recorded wavenumber against its text: 560/560 bands and 60/60 records were confirmed (Wilson 95% CI 99.3–100% and 94–100%). This bounds *transcription* error — hallucinated, mis-parsed or unit-mangled values — below 1% of bands. It does not measure whether the parser found every IR string in every paper — a recall question requiring human reading; that extraction-recall audit is formally deferred (§8).

`irexp_resolved` (43,060 records, 100% structure-linked) is the benchmark-ready split,  $\approx 6\times$  the 6,833-molecule set used to train Spectro<sup>1</sup> (Fig. S2).

### 3. Benchmark design (IRSpectra-Bench)

From `irexp_resolved` we draw *IRSpectra-Bench*, 194 blind elucidation problems, each giving the *molecular formula* (as from HRMS), the *IR band list* and the  $^1\text{H}$  and  $^{13}\text{C}$  *shift lists*; no name, SMILES or scaffold hint. In 10 of the 194 the authors’ peak assignments name a ring system (2CH<sub>2</sub>·pyrrolidine, pyridazinone H5). Those compounds are solved 1/10 against 54/184 (29.3%) for the rest (`scripts/prompt_leakage.py`), so the annotation does not carry the headline; we keep them rather than drop them. Main-round ground truths are spectrally validated by an automated RDKit check ( $^{13}\text{C}$  peaks vs symmetry-unique carbons, formula match, SELFIES round-trip) excluding merged or incomplete spectra — 6/140, leaving 134; `scripts/validate_benchmark.py` applies the same filter to all three rounds, regenerating every `clean_qids.json` from the released questions and answers.

The filter does not gate on  $^1\text{H}$ . In 13 of the 194 retained records the reported  $^1\text{H}$  integral exceeds the reference structure’s hydrogen count — residual solvent, water, exchangeable protons, or a rotamer mixture — printed as a diagnostic, not excluded; the cohort was fixed in advance, and re-filtering post hoc on a criterion chosen after seeing results is a degree of freedom a benchmark should not take. We report the sensitivity instead: dropping all 13 moves the headline from 28.4% to 29.3% (53/181), +0.9 points, leaving every conclusion unchanged. The controlled rounds’ 60 compounds are fixed, pre-registered sets, audited the same way but reported rather than filtered (57/60 pass; §8).

**Difficulty is declared in advance, as a property of the compound.** By RDKit ring analysis a compound is *simple* iff it has at most two rings, no fused/spiro/bridgehead system and  $\leq 22$  heavy atoms; all others are *complex* (98 simple / 96 complex). The threshold is not load-bearing: sweeping it from 18 to 26 leaves the simple-minus-complex top-1 gap inside a 36–40-point band throughout (36.1 at the narrowest, 39.6 as released; `scripts/difficulty_sensitivity.py`). This axis is dis-

tinct from the continuous size gradient of §4.1 ( $\leq 15$  / 16–25 / >25 heavy atoms), and InChIKey de-duplication across rounds prevents leakage.

**The 50/50 balance is by design.** The sampler fills each difficulty stratum to half the round; the eligible corpus is 17.5% simple / 82.5% complex. Reweighted to that composition, top-1 falls 28.4%  $\rightarrow$  15.2% [10.6–20.4] and recall 33.5%  $\rightarrow$  19.8% [14.4–25.8]; verification precision must be reweighted too (89.2%  $\rightarrow$  71.5% [50.2–92.5]). We report the benchmark figure as released and the reweighted figure as the better estimate on an arbitrary paper.

**Scoring is mechanical.** A prediction is *correct* if its RDKit InChIKey connectivity layer (first 14 characters) matches the reference: we score *constitution*, so correct constitution with wrong stereochemistry counts as correct. The *full*, stereochemistry-sensitive InChIKey gives 21.1% top-1 and 25.8% recovered (41/194 and 50/194) against 28.4% / 33.5% — 7.3 points lower on top-1 and 7.7 on recovered — so the constitution figure is an upper bound on full-stereochemistry accuracy. Constitution is the headline because 1D  $^1\text{H}/^{13}\text{C}/\text{IR}$  rarely fixes absolute configuration: only 10.3% (20/194) of answers carry a *defined* (assigned R/S) stereocentre, and a model is penalised there for information the prompt never contained (`scripts/score_main.py --stereo`).

**Metrics.** *Top-1 (exact constitution)*: best-ranked candidate matches at the connectivity layer; *recovered (top-3)*: reference among up to three returned candidates, the lenient “recovery” protocol of ref.<sup>2</sup>. *Generation recall*: reference in the candidate pool *before* re-ranking, the ceiling any verifier can reach (coincides with recovered except in §5.3, where the pool is larger). *Verification precision (conditional on recall)* over recall-positive compounds isolates verification from generation. The forward-verifier ranks by symmetric *chamfer distance* between predicted and observed  $^{13}\text{C}$  peak sets (lower is better); Morgan(2, 2048) Tanimoto<sup>40</sup> gives a graded scaffold-family signal. CIs are bootstrap 95%; model-vs-model differences use McNemar’s exact test<sup>41</sup> with Holm correction<sup>42</sup>.

Solvers are frontier LLM agents (Claude Opus), one sub-agent per batch, closed-book under a consumer subscription: no tools beyond an RDKit formula check, no ground truth, verified by grep-auditing transcripts. Formula-adherence rates by round are reported in the Electronic Supplementary Information.

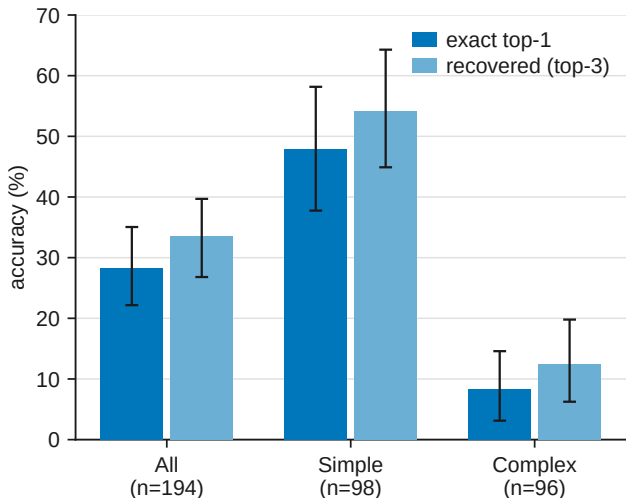
## 4. How well do LLMs elucidate real structures?

### 4.1 Headline performance

Solver agents work blind from formula + IR +  $^1\text{H}$  +  $^{13}\text{C}$ , returning up to three ranked candidates in bounded contexts reset every 2–12 compounds rather than one long context — the

variable §4.3 shows what matters. The benchmark comprises all 194 compounds (134 spectrally-validated + 60 controlled-round; data/benchmark\_main/raw/).

A strictly-validated subset (134 main-clean + 57 controlled-clean = 191) reproduces this within  $\approx 1$  point on every metric — top-1 28.8%, 95% CI 23–36; recall 34.0%; simple 49.0% / complex 8.4% — so the asymmetric inclusion of pre-registered controlled sets does not drive it. The deliberate 50/50 difficulty balance does: reweighted to the 17.5%-simple composition of the eligible corpus, top-1 is 15.2% [10.6–20.4] and recall 19.8% [14.4–25.8] (§3). Per-stratum figures are unchanged; the reweighted number is what to read as performance on an arbitrary paper. Accuracy falls monotonically with size (Fig. 2): 60.5% top-1 at  $\leq 15$  heavy atoms, 28.3% at 16–25, 7.0% above 25 (Fig. S3).



**Fig. 2.** Top-1 and recovered accuracy on IRSpectra-Bench by difficulty (all / simple / complex, n=194), with bootstrap 95% confidence intervals. The benchmark separates a realistic difficulty range: simple targets are solved four to six times as often as complex ones on both metrics.

Of the 137 analysable top-1 misses (139 in all; two predictions did not parse; scripts/analyze\_misses.py), 76.6% are constitutional isomers of the true structure against 23.4% with the wrong formula — not mostly regiochemistry: only 22.6% share the true Murcko scaffold, 2.9% reach Tanimoto  $\geq 0.85$ , and the median isomeric-miss Tanimoto is 0.39. Strict regiochemistry accounts for a fifth of failures; most misses are larger rearrangements of the same atoms. Near-degenerate  $^1\text{H}/^{13}\text{C}$  shifts under-determine connectivity; the misses show *wrong connectivity at the right composition*.

#### 4.2 Reconciling with prior reports

Our 28% top-1 sits far below curated NMR-only demos<sup>2</sup> and below in-distribution trained baselines (48–94%)<sup>1,8,20</sup>: difficulty, scoring leniency, starting-material hints and hand curation inflate those reports, while settings differ on every axis from our blind literature spectra. The  $\approx 40\times$  MolPuzzle swing for one model (§1.1) bounds what unaudited near-100% claims can bear.

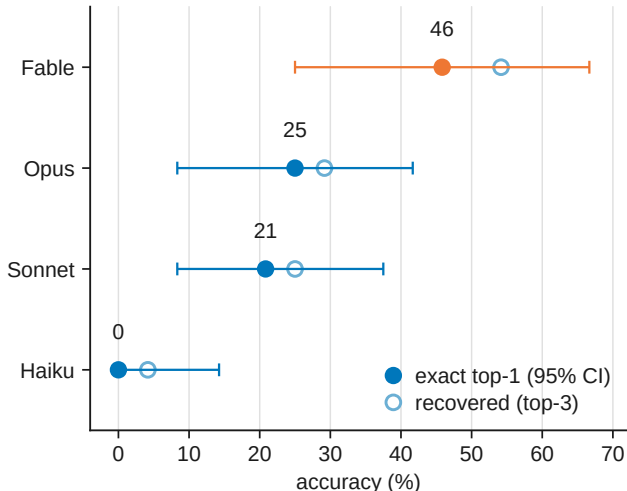
#### 4.3 Methodology dominates: a within-compound control

On the same 20 molecules, recovered rose 5%  $\rightarrow$  15% (and top-1 0%  $\rightarrow$  15%; McNemar  $p=0.25$ ) when four bounded, tool-using agents replaced one long context — directional only.

#### 4.4 Model comparison: the benchmark orders capability but separates only the extremes

We solved a fixed 24-compound subset blind with four Claude models spanning a wide capability range, including the newest (Table 5), under one prompt, one scorer and one candidate budget (Fig. 3, Table 3).

One protocol asymmetry must be disclosed: the three comparison models solved the 24 compounds as four six-compound contexts each, whereas the Opus column reuses the headline run, where those items sat in one six-compound and two twelve-compound contexts — the variable that §4.3 measures at 5%  $\rightarrow$  15%. The Opus estimate is therefore not protocol-matched and, if anything, handicapped by the longer contexts; this touches neither the nesting nor the Fable-vs-Haiku contrast, and a clean re-run is outstanding in docs/MODELS.md.



**Fig. 3.** Four-model comparison on a fixed 24-compound subset, solved blind under one protocol. Outcomes are strictly nested — each stronger model solves a superset of the weaker one’s compounds — so the benchmark is capability-sensitive, but at n=24 it is underpowered to separate adjacent models (§4.4).

CIs are bootstrap except Haiku’s: Clopper–Pearson exact for 0/24, where the percentile bootstrap is degenerate.

Outcomes are strictly nested (Haiku  $\subset$  Sonnet  $\subset$  Opus  $\subset$  Fable); only Fable-vs-Haiku reaches significance (Holm  $p=0.006$ ). Opus headline runs used longer contexts than the comparison models (§4.3).

#### 4.5 Domain case study: battery-electrolyte chemistry

IRSpectra-Bench-Electrolyte (46 scored compounds across six electrolyte functional classes, held out and solved blind) matches the headline regime: top-1 26%, recovered 28%; the true structure enters the pool for 13/46 and ranks first in 12/13 — recall-bound, not domain-specific (Fig. S4, Table 4). Per-class intervals overlap ( $\chi^2$   $p\approx 0.56$ );  $\text{sp}^3\text{-C-F}$  reaches 50%, sulfonyl and nitrile 12%.

Intervals overlap throughout (six-class  $\chi^2$   $p\approx 0.56$ ; best-vs-worst Fisher exact  $p\approx 0.28$ ), so the ordering is hypothesis-generating, not established — C–F couplings pin  $\text{sp}^3\text{-C-F}$  regiochemistry, while sulfonyl and nitrile targets turn on oxidation-state ambiguity and heteroaromatic substitution that  $^1\text{H}/^{13}\text{C}$  shifts underdetermine. It is the recall-bound failure of §4.1 in a domain-specific guise, and it is where §5’s *forward-*

**Table 2. Headline elucidation performance on IRSpectra-Bench (n=194).** Overall and by difficulty stratum, with bootstrap 95% CIs.

| metric                                      | overall (n=194)      | simple (n=98) | complex (n=96) |
|---------------------------------------------|----------------------|---------------|----------------|
| top-1 exact constitution                    | <b>28.4%</b> [22–35] | 48.0% [39–57] | 8.3% [3–15]    |
| recovered (top-3)                           | 33.5% [27–40]        | 54.1% [44–63] | 12.5% [6–20]   |
| scaffold-level (best Tanimoto $\geq 0.45$ ) | 56%                  | 73%           | 39%            |
| mean best Tanimoto                          | 0.59                 | 0.73          | 0.45           |

**Table 3. Four-model comparison on a fixed 24-compound subset.** All four models ran the identical blind protocol.

| model          | top-1      | recovered | top-1 95% CI |
|----------------|------------|-----------|--------------|
| Claude Fable 5 | <b>46%</b> | 54%       | [25–67]      |
| Claude Opus    | 25%        | 29%       | [8–42]       |
| Claude Sonnet  | 21%        | 25%       | [8–38]       |
| Claude Haiku   | 0%         | 4%        | [0–14]       |

verification recipe plays a role *analogous* to (not a replacement for) computational-NMR validation.

#### 4.6 Is the model reading the spectra? A formula-only control

We reran the blind protocol with spectra masked, leaving only the formula (<sup>43</sup>).

Outcomes are perfectly nested (McNemar  $p=0.001$ ). Accuracy is flat in source-paper year across all 194 ( $r=-0.007$ ; panel b of Fig. 4), bounding pretraining recall. We claim a strong bound, not exclusion (§8).

#### 4.7 Does the diagnosis hold outside one vendor? A four-vendor replication

Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol solved the same 60-compound arm under the identical protocol (Table 6). Verification precision exceeds generation recall in every arm (panels c–d of Fig. 4); bootstrapping separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok’s gap (+9.2) is directional. Three models beat Claude on recall; candidate budgets differ (ours 2.20 vs 3.00 per compound), so recall rankings are approximate. A clean-clone control for Grok (recall 28/60 vs 32/60, McNemar  $p=0.39$ ) bounds key leakage. Models ran through a coding-assistant harness with undisclosed reasoning tiers (§8).

## 5. Forward-verification elucidation

### 5.1 Method

The inverse direction is the model’s hard, isomer-blind direction; the forward direction (structure  $\rightarrow$  spectrum) is its easy, accurate one<sup>2</sup>. We close a training-free generator–verifier loop:

generate candidate structures (inverse)  $\rightarrow$   
forward-predict each candidate’s <sup>13</sup>C spectrum,  
blind to the observed spectrum  $\rightarrow$  re-rank can-

didates by the distance between predicted and observed <sup>13</sup>C  $\rightarrow$  return the best match.

The inverse solver’s 373 deduplicated candidates across 194 targets are forward-predicted in shuffled, anonymised batches, blind to the observed spectrum and target identity. Predicted and observed <sup>13</sup>C peak sets are compared by symmetric chamfer distance (Fig. 5).

Regioisomers separate by a median 1.21 ppm in forward prediction, but 82% lie within the predictor’s  $\approx 2$  ppm error — a thin margin confirmed against a derangement null at  $p=0.001$  (§5.5). Sharper predictors (GNN, DFT-coupled models<sup>[44,45,46]</sup>) would move the verification ceiling, not search.

### 5.2 Result

The first column below is the 60-compound arm (v3 + v2-control) on which §5.3–§5.5 build; the second, the full benchmark. All 373 candidates were forward-predicted — nothing here is a lower bound; the self-ranking row re-derives Table 2 from §4’s output.

When the true structure is among the candidates, forward-verification selects it in 58 of 65 cases (89%). Of the 65, 28 had a single candidate — scored by construction by any ranker — so the pooled figure is not the verifier’s margin over chance. Where a choice existed ( $n=37$ ), forward-verification gets 30/37 (81%) against a 54.0% derangement floor (§5.5) — +27.1 points; self-ranking 27/37 (73%).

The margin over self-ranking remains small and unresolved: seven compounds gained, four lost, McNemar exact  $p=0.55$ . We do not claim it. Precision is high in absolute terms while recall binds: top-1 moves only 28%  $\rightarrow$  30% because the true structure was never proposed for 129 of 194 compounds, which no re-ranking can repair. The verifier converts recall almost completely on *simple* targets (50/53, 94%) and ties self-

**Table 4. Per-class performance on IRSpectra-Bench-Electrolyte (n=46).**

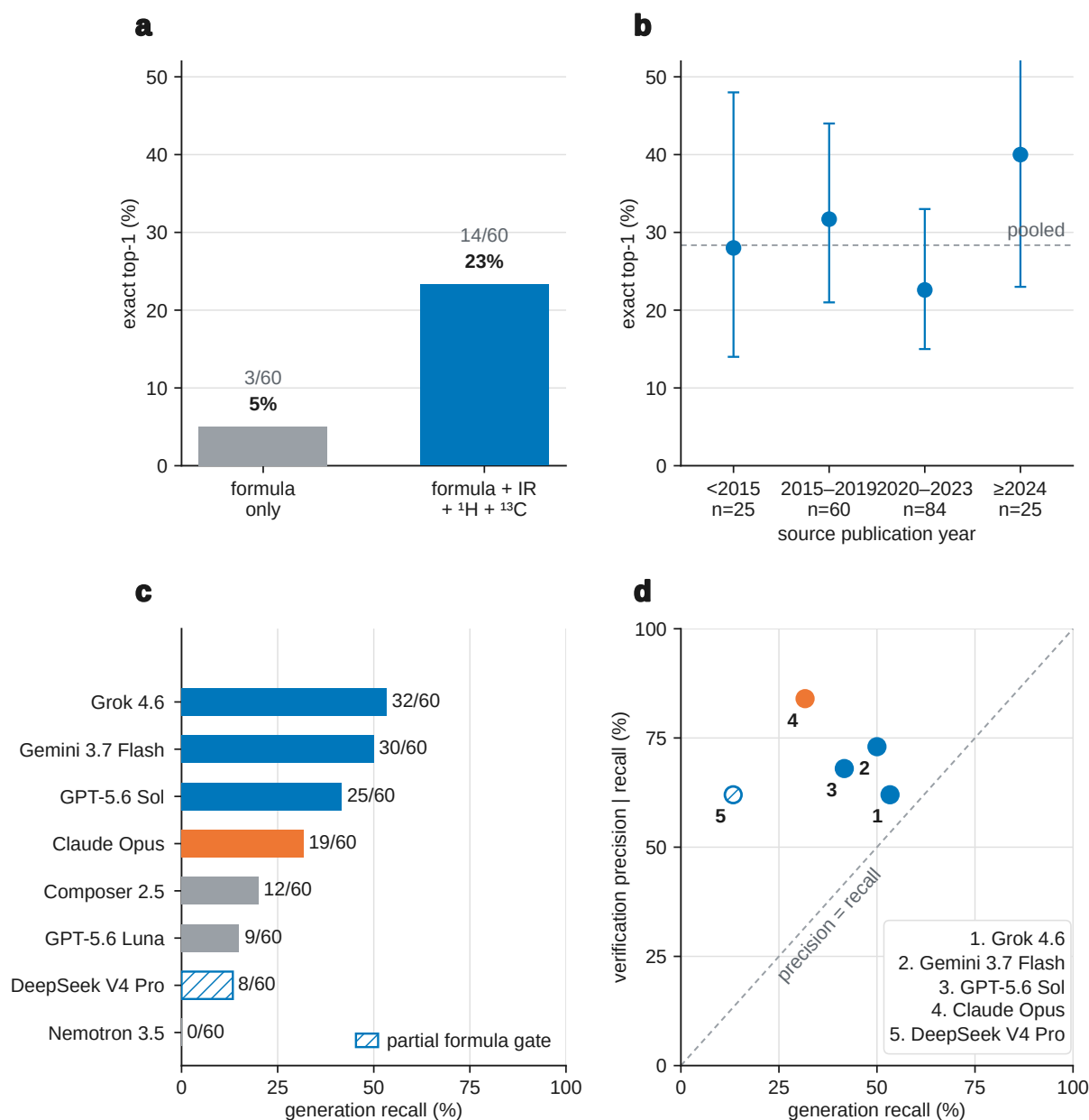
| electrolyte class    | n | top-1      | 95% CI  | recovered (top-3) | 95% CI  |
|----------------------|---|------------|---------|-------------------|---------|
| sp <sup>3</sup> -C–F | 8 | <b>50%</b> | [22–78] | 50%               | [22–78] |
| carbonate            | 7 | 29%        | [8–64]  | 43%               | [16–75] |
| phosphoryl           | 8 | 25%        | [7–59]  | 25%               | [7–59]  |
| glyme / oligoether   | 7 | 29%        | [8–64]  | 29%               | [8–64]  |
| sulfonyl / sulfonate | 8 | 12%        | [2–47]  | 12%               | [2–47]  |
| nitrile              | 8 | 12%        | [2–47]  | 12%               | [2–47]  |

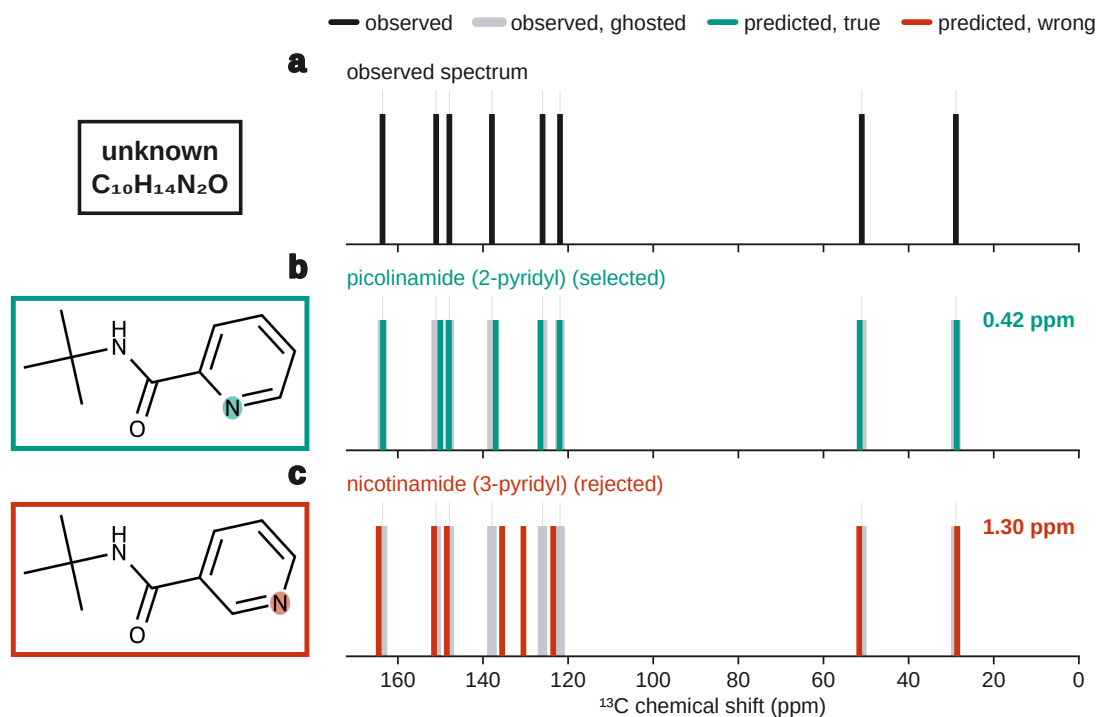
**Table 5. Formula-only control.** Paired on the same 60 compounds as §5.

| condition                                     | top-1            | recovered (top-3) |
|-----------------------------------------------|------------------|-------------------|
| formula only                                  | <b>3/60 (5%)</b> | 3/60 (5%)         |
| formula + IR + $^1\text{H}$ + $^{13}\text{C}$ | 14/60 (23%)      | 19/60 (32%)       |

**Table 6. Cross-vendor decomposition on the 60-compound arm.** Recall and precision have different denominators, so the criterion is the inequality rather than a difference.

| model                 | generation recall   | verification precision   recall | multi-candidate only |
|-----------------------|---------------------|---------------------------------|----------------------|
| Claude Opus (Table 7) | 19/60 = 32% [21–44] | 16/19 = 84% [62–94]             | 10/13 = 77%          |
| Grok 4.6              | 32/60 = 53% [41–65] | 20/32 = 62% [45–77]             | 20/32 = 62%          |
| Gemini 3.7 Flash      | 30/60 = 50% [38–62] | 22/30 = 73% [56–86]             | 22/30 = 73%          |
| GPT-5.6 Sol           | 25/60 = 42% [30–54] | 17/25 = 68% [48–83]             | 16/24 = 67%          |

**Fig. 4.** Robustness of the recall-bound diagnosis. **(a)** Formula-only vs full modality on 60 compounds. **(b)** Accuracy vs source-paper year ( $n=194$ ; point-biserial  $r=-0.007$ ; dashed line = pooled 28%). **(c)** Generation recall by model on the 60-compound arm (orange = Claude Opus headline; grey = below formula-adherence gate). **(d)** Verification precision vs generation recall (numbered key; §4.7).



**Fig. 5.** Forward-verification on a benchmark regioisomer pair: picolinamide and nicotinamide are indistinguishable to the inverse task, but forward-predicted  $^{13}\text{C}$  sticks separate the true isomer from the alternative (§5.1).

**Table 7. Forward-verification decomposition.** Original arm and full benchmark.

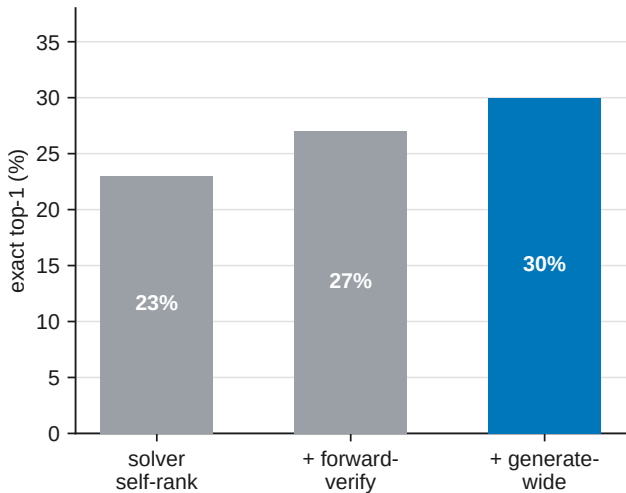
|                                                     | 60-compound arm | full benchmark (n=194) |
|-----------------------------------------------------|-----------------|------------------------|
| generation recall                                   | 19/60 (32%)     | 65/194 (34%)           |
| top-1, solver self-ranking                          | 14/60 (23%)     | 55/194 (28%)           |
| top-1, forward-verified re-ranking                  | 16/60 (27%)     | 58/194 (30%)           |
| <i>conditional on recall</i> — self-ranking         | 14/19 (74%)     | 55/65 (85%)            |
| <i>conditional on recall</i> — forward-verification | 16/19 (84%)     | 58/65 ( <b>89%</b> )   |
| ...single-candidate compounds within that set       | 6/19            | 28/65                  |
| <i>multi-candidate only</i> — self-ranking          | 8/13 (62%)      | 27/37 (73%)            |
| <i>multi-candidate only</i> — forward-verification  | 10/13 (77%)     | 30/37 (81%)            |

ranking on *complex* ones (8/12, 67%). Elucidation factorises into two near-independent levers: a strong verifier and a generator at 34% recall that is the wall (Fig. 1).

### 5.3 Generate-wide: testing the recipe

The decomposition implies *generate wide, verify by forward prediction* — a chemistry analog of self-consistency sampling<sup>47</sup>. Ten solver agents proposed up to six regiochemistry-aware candidates per compound, pooled with the originals and re-ranked.

Wide generation lifts recall 32%  $\rightarrow$  42% and top-1 23%  $\rightarrow$  30% (McNemar  $p=0.34$ ; Fig. 6). Recall plateaus at 42% on polycyclic targets; verification precision falls 84%  $\rightarrow$  72% — the training-free ceiling. Roughly a third of misses need only regiochemistry around a correct scaffold; two thirds need a scaffold never proposed.



**Fig. 6.** Forward-verification inference ladder on the 60-compound arm (§5.3). Each stage adds a check on the same compounds; the hero bar is generate-wide top-1.

### 5.4 Non-LLM verifiers: a deterministic lookup and a learned model

§5.3 suggests replacing the verifier with a non-LLM  $^{13}\text{C}$  predictor. We tested a HOSE-code<sup>48</sup>-style lookup and a small message-passing GNN, both trained on the same nmrshiftdb2 dump<sup>49</sup> and applied to the same §5.2 candidate sets so only the predictor changes. The GNN is deliberately modest; sharper purpose-built  $^{13}\text{C}$  models exist<sup>44,45,50</sup>.

The HOSE lookup ties self-ranking; the GNN reaches 59/65 (91%) vs the LLM’s 58/65 (89%), directionally (Table 9, Fig. S6). Wagen’s MagNET-in-the-loop study<sup>51–53</sup> moves connectivity 46.4%  $\rightarrow$  55.4% on eight problems — same order as our ladder, underpowered at  $n=8$ .

### 5.5 Negative control

Deranging observed  $^{13}\text{C}$  pairings collapses verification precision 89.2%  $\rightarrow$  73.8% mean (one-sided  $p=0.001$ ); on multi-candidate sets alone, 81.1% vs a 54.0% floor (+27.1 points). Chamfer margin does not support selective prediction (flat top-1 across coverage fractions) — a reported null.

### 5.6 Is the recall wall task-intrinsic? A trained-generator probe

That ceiling is a property of training-free LLM elicitation, not the task. Pooling a small IReXP-fine-tuned  $^1\text{H}/^{13}\text{C} \rightarrow \text{SMILES}$  generator with Claude’s candidates lifts recall 33.5%  $\rightarrow$  54.1% and top-1 28.4%  $\rightarrow$  35.1% (McNemar  $p=0.015$ ; Fig. S5). Zero-shot without IReXP fine-tuning recovers 0/248; with it, 25%. NMR-Solver (52.9%)<sup>27</sup> and trained IR transformers (63.8%)<sup>20</sup> confirm the task is not bound at 28–30%.

## 6. The decomposition across the published literature

The split measured above applies to any system reporting top-1 =  $a$  and top- $k$  =  $b$ : recall  $\geq b$ , conditional precision  $\leq a/b$ . Table 10 applies this to every system we could read in full (scripts/literature\_decomposition.py).

**Three groups changed only the data** and recall carried 68% (NMR-Solver simulated  $\rightarrow$  real), 70% (SpecX random  $\rightarrow$  scaffold) and 83% (Espejo education  $\rightarrow$  industrial) of each collapse. Published gains come mainly from candidate supply (NMRAgent ablation<sup>28</sup>, NMRGym formula ablation<sup>5</sup>). Recall binds where spectra are real and heterogeneous (87–91% of our loss); ranking dominates on simulated or single-library data (38–39%). The field almost never reports whether the true structure was proposed at all (<sup>18</sup> excepted).

## 7. Discussion

Frontier LLMs are good verifiers (89% conditional precision) and weak proposers (34% recall) on real literature spectra. The contribution is a diagnosis with a bounded, training-free improvement, not a solved elucidator (§8).

That split held across four Claude models, a battery-electrolyte subset, four verifiers and four vendor families (§6); concurrent systems that move the same levers are surveyed in §1.1.

Training is not foreclosed: IReXP fine-tuning lifts recall to 54% (§5.6). What compounds is open experimental data, honest benchmarks, and inference scaffolding that rides each new model. Two practical findings: bounded contexts with tool access may help (§4.3); use forward-predicted  $^{13}\text{C}$  agreement to re-rank, not as an abstention gauge (§5.5).

## 8. Limitations

Scoring is mechanical; solver runs were transcript-audited for zero ground-truth access (transcripts on request). **(i) Consumer harness.** Runs used a consumer subscription that exposes no model snapshot, temperature, seed or thinking tier; exact inference is not reproducible. Scoring is: frozen predictions, ground truth and mechanical scorers regenerate every training-free number. The instruction text wrapping Claude solver and forward-prediction batches was not captured — only per-compound payloads are released — so the verbatim prompts in the ESI are the cross-vendor harness prompts. **(ii) Pretraining contamination** is bounded by formula-only (23%  $\rightarrow$  5%) and flat recency controls (§4.6) but not excluded; verbatim spectral strings from PMC in the prompt remain a retrieval confound. **(iii) Object type, formula and scoring.** Inputs are author-transcribed band and shift lists, not raw absorbance traces or FIDs — a different object from digitised spectra and from Espejo *et al.*<sup>30</sup>. The molecular formula is supplied (as from HRMS); systems such as NMIRacle<sup>8</sup> that

**Table 8. Generate-wide vs original.** On the 60-compound arm only, since wide generation was run there; the “original” column is Table 7’s first.

|                                                     | original (60-compound arm) | generate-wide |
|-----------------------------------------------------|----------------------------|---------------|
| generation recall (true structure among candidates) | 32%                        | <b>42%</b>    |
| forward-verified top-1                              | 27%                        | 30%           |
| verification precision (conditional on recall)      | 84%                        | 72%           |

**Table 9. Verifier comparison, conditional on recall.**

| verifier                         | 60-compound arm (n=19) | full benchmark (n=65) | held-out <sup>13</sup> C MAE |
|----------------------------------|------------------------|-----------------------|------------------------------|
| solver self-ranking              | 14/19 (74%)            | 55/65 (85%)           | —                            |
| deterministic HOSE <i>lookup</i> | 14/19 (74%)            | 55/65 (85%)           | 3.23 ppm                     |
| learned GNN (same data)          | 16/19 (84%)            | <b>59/65 (91%)</b>    | 1.70 ppm                     |
| LLM forward-verifier (§5.2)      | 16/19 (84%)            | 58/65 (89%)           | —                            |

**Table 10.** Recall and ranking loss from published top-*k* figures. Rows are comparable only within a correctness criterion; stereochemistry-strict and connectivity figures differ by 21 points on NMR-Solver’s identical predictions. Candidate budgets are given because recall over three candidates is mechanically smaller than over ten.

| system                                         | data (n)           | top-1 | top- <i>k</i> ( <i>k</i> ) | recall | prec.        |
|------------------------------------------------|--------------------|-------|----------------------------|--------|--------------|
| <b>scored on connectivity</b>                  |                    |       |                            |        |              |
| this work, solver alone                        | literature (194)   | 28.4% | 33.5% (3)                  | 33.5%  | 84.6%        |
| this work, + forward-verification              | literature (194)   | 29.9% | 33.5% (3)                  | 33.5%  | <b>89.2%</b> |
| NMR-Solver <sup>27</sup>                       | literature (450)   | 52.9% | 67.3% (10)                 | ≥67.3% | ≤78.6%       |
| NMRAgent <sup>28</sup>                         | literature (450)   | 61.6% | 70.0% (10)                 | ≥70.0% | ≤88.0%       |
| <b>scored with stereochemistry</b>             |                    |       |                            |        |              |
| this work                                      | literature (194)   | 21.1% | 25.8% (3)                  | ≥25.8% | ≤81.8%       |
| NMR-Solver <sup>27</sup>                       | simulated (1,000)  | 66.9% | 89.9% (10)                 | ≥89.9% | ≤74.4%       |
| NMR-Solver <sup>27</sup>                       | literature (450)   | 31.6% | 53.8% (10)                 | ≥53.8% | ≤58.7%       |
| Espejo Morales <sup>30</sup>                   | education (236)    | 80.9% | 90.0% (5)                  | ≥90.0% | ≤89.9%       |
| Espejo Morales <sup>30</sup>                   | industrial (34)    | 20.6% | 29.1% (5)                  | ≥29.1% | ≤70.9%       |
| <b>exact match, stereo handling not stated</b> |                    |       |                            |        |              |
| Alberts, 6–13 atoms <sup>20</sup>              | NIST IR (3,455)    | 63.8% | 84.0% (10)                 | 84.0%  | 75.9%        |
| Alberts, 5–35 atoms <sup>20</sup>              | NIST IR (5,024)    | 59.9% | 78.5% (10)                 | 78.5%  | 76.4%        |
| SpecX, random split <sup>4</sup>               | simulated (99,439) | 59.0% | 81.8% (10)                 | 81.8%  | 72.2%        |
| SpecX, scaffold split <sup>4</sup>             | simulated (99,439) | 29.7% | 50.6% (10)                 | 50.6%  | 58.7%        |
| IR-Agent <sup>17</sup>                         | NIST IR (905)      | 10.3% | 21.6% (10)                 | 21.6%  | 47.7%        |
| <b>criterion not stated</b>                    |                    |       |                            |        |              |
| Priessner <sup>18</sup>                        | experimental (34)  | 20.6% | —                          | 26.5%  | 77.8%        |

take no formula solve a harder prior, so absolute accuracies are not interchangeable. Headline metrics score constitution (InChIKey connectivity); with stereochemistry, top-1 is 21.1%. **(iv) Statistical honesty.** Forward-verification vs self-ranking is unresolved (McNemar  $p=0.55$ ); generate-wide top-1 like-wise ( $p=0.34$ ). The four-model comparison at  $n=24$  is under-powered for adjacent ranks and carries a disclosed protocol asymmetry (§4.4). The inference ladder diagnoses a bottleneck; it is not a demonstrated accuracy advance. **(v) Missing on-bench baselines.** Spectro, NMIRacle and CASE were not scored on IRSpectra-Bench, so the LLM recall wall is not cleanly separated from harness or modality choice — left to future leaderboard work. **(vi) Cross-vendor scope.** Headline  $n=194$  is Claude Opus; the four-vendor replication is on 60 compounds. Candidate budgets differ across vendors (Claude mean 2.20 vs 3.00), so recall rankings are approximate (§4.7). **(vii) Deferred and abandoned arms.** The expert-chemist audit of elucidation outputs is frozen at `data/audit/` and formally deferred — not run. Leave-one-modality-out ablation (noIR/noH/noC) was specified and never completed; it is abandoned for this manuscript (`docs/MODELS.md`; ESI). The extraction-recall human audit of parser coverage (§2.3) is likewise deferred. **(viii) Scope.** Battery subset uses literature electrolyte chemistry, not operando spectra. Single-sample scoring per compound (bootstrap CIs reflect compound sampling only). Organic literature bias of PMC-OA sources.

## 9. Methods

**Mining and resolution.** PMC-OA full text was fetched from `s3://pmc-oa-opendata`, parsed deterministically, and resolved with OPSIN, RDKit and SELFIES, with an optional cached PubChem fallback.

**Benchmark and agents.** Problems were sampled from `irexp_resolved`, stratified by RDKit ring analysis, and de-duplicated across rounds by InChIKey. A compound was admitted only if its raw  $^1\text{H}$  payload — multiplicities and  $J$  values as printed — could be re-extracted verbatim from the PMC-OA full text of its source article (`benchmark_v2.raw_1h_for`), which puts genuine coupling information in the prompt and, unavoidably, the source article’s exact string (§8). The battery-electrolyte subset (§4.5) was drawn by SMARTS filters for six electrolyte functional classes (carbonate, sulfonyl/sulfonate, nitrile,  $\text{sp}^3\text{-C-F}$ , phosphoryl, glyme/oligoether), balanced to eight compounds per class (48 curated; 46 scored after two yielded no parseable candidate), excluding compounds used elsewhere; it was  $J$ -enriched and spectrally validated. Solver and forward-prediction agents were independent Claude-Opus sub-agents under the same subscription, instructed closed-book and audited for zero web/answer access. Scoring used RDKit InChIKey-connectivity match; similarity Morgan(2, 2048) Tanimoto; forward-verification a symmetric chamfer distance over  $^{13}\text{C}$  peak sets. The core protocol trains no model and uses no paid API; the \$5.6 generator and the \$5.4 learned  $^{13}\text{C}$  verifier are the only trained components, reported as complements and fenced from the headline results.

**Models and versions.** Headline runs used Claude Opus via consumer subscription (2026-06-09–11); formula-only control 2026-07-28; forward-prediction extension 2026-08-07. No model snapshot or decoding parameters can be reported — the harness exposes neither (§9). Fixed are frozen per-compound outputs and mechanical scorers (`docs/MODELS.md`).

**Reproducibility.** Every round is frozen: questions, ground-truth answers, per-agent raw outputs, predictions and scorer outputs are released; the sampler, scorer and forward-verification harness are scripted end-to-end.

## Supporting Information

Supplementary figures and extended methods appear in the Electronic Supplementary Information (`docs/paper_esi.pdf`).

## Author contributions

**I.Y.:** conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing — original draft.

**R.S.:** methodology, software, formal analysis, investigation, validation (trained-generator and learned-verifier probes, \$5.4 and \$5.6), writing — review and editing.

**R.A.V.-H.:** conceptualization, methodology, supervision, writing — review and editing.

## Conflicts of interest

There are no conflicts to declare.

## Data availability

Data, predictions, scoring scripts and figure regeneration are in the project repository; Table 11 lists each component. A frozen snapshot will be deposited on Zenodo (DOI at proof); GitHub is the development mirror. IRExp is also mirrored at [huggingface.co/datasets/ilkhamfy/IRExp](https://huggingface.co/datasets/ilkhamfy/IRExp) (use `data/train_no_bench.jsonl.gz` for fine-tuning without benchmark leakage).

IRExp redistributes extracted numeric data only (band lists, shifts, structures, source DOIs) from PMC-OA (CC-BY-4.0) and Chemotion (CC-BY-SA-4.0); pools are separable by `source_doi` (`scripts/split_license_pools.py`). Code is MIT. Cite the Zenodo deposit and attribute sources via each record’s `source_doi`. De-leak with `contrib/generator_probe/build_exp_manifest.py` before training against the benchmark; withhold `data/audit/key.jsonl` and `data/modality/key.json` from blinded reviewers.

**Table 11. Headline released artefacts.** Script-level inventory lives in the repository README.

| component                                     | data and code                                                                                                    |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| IRexp / irexp_resolved                        | data/irexp/, data/irexp_resolved/                                                                                |
| benchmark + within-compound control           | data/benchmark*/ — scripts/benchmark_v2.py                                                                       |
| headline scoring (Table 2)                    | scripts/score_main.py                                                                                            |
| forward-verification (+ full-bench extension) | data/fverify/, data/fverify_main/ — scripts/forward_verify.py, scripts/forward_verify_main.py                    |
| generate-wide (§5.3)                          | data/gw/, data/fverify2/, data/fverify_gw/ — scripts/score_generate_wide.py                                      |
| cross-vendor (§4.7)                           | data/cross_vendor/ — scripts/cross_vendor_budget.py, scripts/cross_vendor_gap.py                                 |
| contamination / recency (§4.6)                | data/modality/, data/audit/recency_control.json — scripts/modality_ablation.py, scripts/contamination_recency.py |
| figures                                       | docs/figures/ — scripts/make_all_figures.sh                                                                      |
| trained generator + GNN verifier              | contrib/generator_probe/, data/fverify_gen/ — scripts/gnn_predict.py, data/nmrshiftdb/gnn_c13.pt                 |
| integrity gates                               | scripts/check_manuscript.py, scripts/check_layout.py                                                             |

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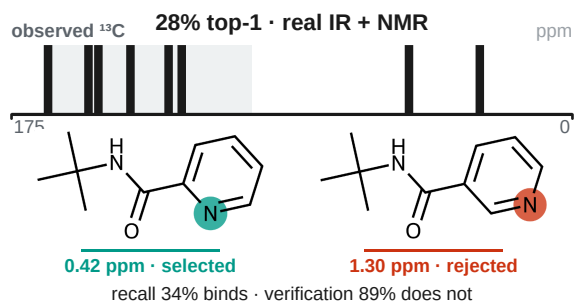
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## Table of contents entry



IRSpectra-Bench and IReXP on real literature IR +  $^1\text{H}$  +  $^{13}\text{C}$ :  
candidate recall, not verification, limits LLM elucidation —  
the true structure is proposed for 34% of 194 compounds and,  
once proposed, selected 89% of the time, lifting top-1 from  
28% to 30%.